

CREATING AN OPTICAL CHARACTER RECOGNITION MODEL USING THE EMNIST DATASET

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PROBLEM

- Schools and big companies use tons of paper every year
 - 420 million tons of paper consumed every year [1]
- Deforestation
 - Hurting the environment
- Digitalization
 - Reducing the need for paper
- Need to convert paper media to digital, editable text



Example of Deforestation [2]

BACKGROUND

- The EMNIST dataset was created in 2017 by Cohen et al. [3]
- Many people have used the dataset to create character recognition models, as well as programs to digitize text.
- However, there is a gap in research when it comes to saving this data in a usable file format.

OTHER SOLUTIONS

- Google Lens, Apple Live Text
 - Uses CNNs trained on the EMNIST dataset to extract text from images
 - Only allows for the copying of text, doesn't maintain format
- Online text-to-PDF generators [4]
 - Perform the same function as already stated applications
 - Copying/pasting text limits the ability to maintain format

NOVELTY, IMPACT, AND WHY MINE IS BETTER

My research is new because I will be incorporating a number of new features into my program, including but not limited to:

- Text Organization Recognition
- Maintaining Format
- Contextual identification

This will allow my program to effectively and efficiently digitize files.

METHODS

- Used the ByMerge EMNIST dataset
 - 814,255 images, 47 unbalanced classes
 - Better mimics the proportion of letters used in the alphabet
 - Contains merge cases for similar upper/lowercase letters (e.g c & C)
- Split the dataset into training and testing splits

```
def load_emnist():  
    train_images = '/content/drive/MyDrive/SRL: Using an EMNIST-Trained Model to Convert Written Text to PDF - Bennett DiCarlo/gzip/gzip/emnist-bymerge-train-images-idx3-ubyte.gz'  
    train_labels = '/content/drive/MyDrive/SRL: Using an EMNIST-Trained Model to Convert Written Text to PDF - Bennett DiCarlo/gzip/gzip/emnist-bymerge-train-labels-idx1-ubyte.gz'  
    test_images = '/content/drive/MyDrive/SRL: Using an EMNIST-Trained Model to Convert Written Text to PDF - Bennett DiCarlo/gzip/gzip/emnist-bymerge-test-images-idx3-ubyte.gz'  
    test_labels = '/content/drive/MyDrive/SRL: Using an EMNIST-Trained Model to Convert Written Text to PDF - Bennett DiCarlo/gzip/gzip/emnist-bymerge-test-labels-idx1-ubyte.gz'  
    train_X = read_idx(train_images)  
    train_Y = read_idx(train_labels)  
    test_X = read_idx(test_images)  
    test_Y = read_idx(test_labels)  
  
    return (train_X, train_Y, test_X, test_Y)
```

```
raw_train_X, raw_train_Y, raw_test_X, raw_test_Y = load_emnist()
```

```
Processing data from /content/drive/MyDrive/SRL: Using an EMNIST-Trained Model to Convert Written Text to PDF - Bennett DiCarlo/gzip/gzip/emnist-bymerge-train-images-idx3-ubyte.gz.  
Dimensions: 3  
Shape: (697932, 28, 28)  
Processing data from /content/drive/MyDrive/SRL: Using an EMNIST-Trained Model to Convert Written Text to PDF - Bennett DiCarlo/gzip/gzip/emnist-bymerge-train-labels-idx1-ubyte.gz.  
Dimensions: 1  
Shape: (697932,)  
Processing data from /content/drive/MyDrive/SRL: Using an EMNIST-Trained Model to Convert Written Text to PDF - Bennett DiCarlo/gzip/gzip/emnist-bymerge-test-images-idx3-ubyte.gz.  
Dimensions: 3  
Shape: (116323, 28, 28)  
Processing data from /content/drive/MyDrive/SRL: Using an EMNIST-Trained Model to Convert Written Text to PDF - Bennett DiCarlo/gzip/gzip/emnist-bymerge-test-labels-idx1-ubyte.gz.  
Dimensions: 1  
Shape: (116323,)
```

Photo of code from data processing[5]

METHODS (CONT.)

- Created Neural Network to train on the training data
- Convolutional Neural Network (CNN)
 - Used in many image identification programs
 - Identifies features such as edges, textures
 - Reduces computational time by normalizing and pooling
- Used testing dataset to evaluate model performance

METHODS (CNN VISUALIZED)

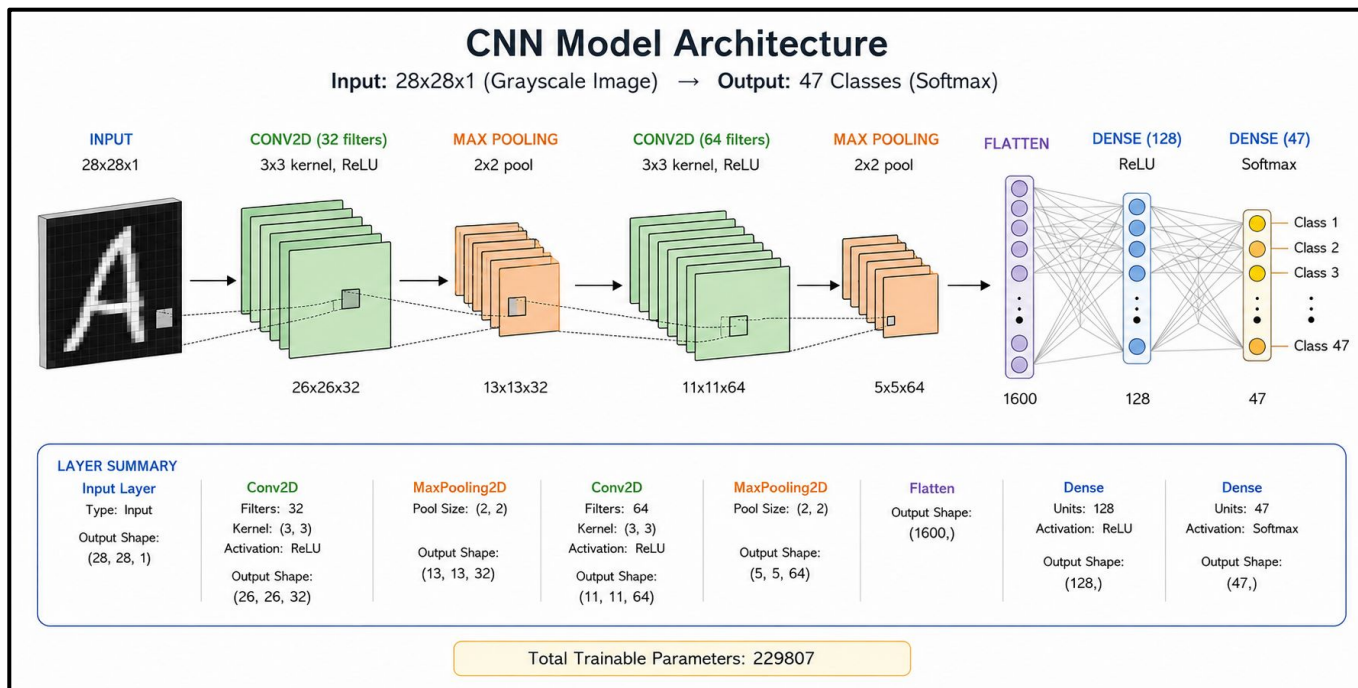


Diagram of Convolutional Neural Network [5]

METHODS (CONT.)

- Created program that identifies contours in an image
- Uses these contours to create bounding boxes
- Cross-references the coordinates of the boxes to identify which lines characters belong to
- Bounding boxes are then manipulated slightly before getting fed into model for identification
- After characters are identified, they are arranged in order, spaces are identified, and all is printed out

METHODS (CONT.)

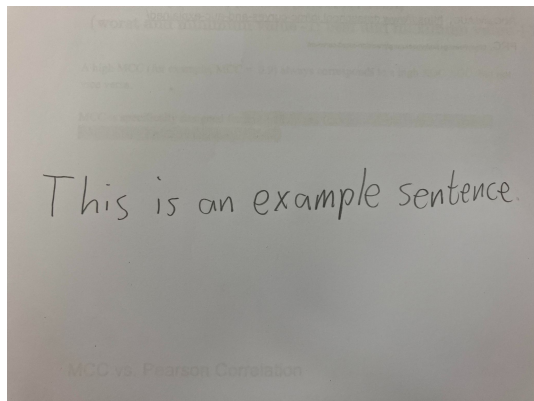


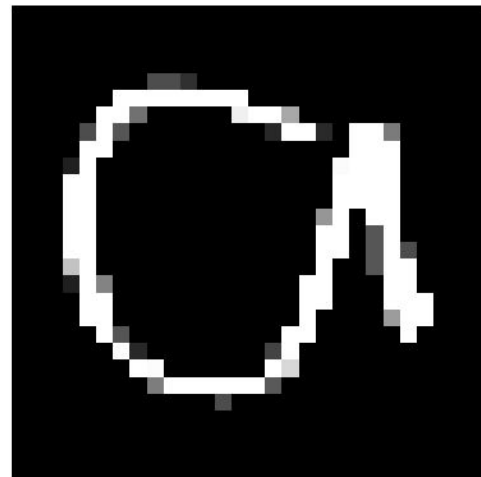
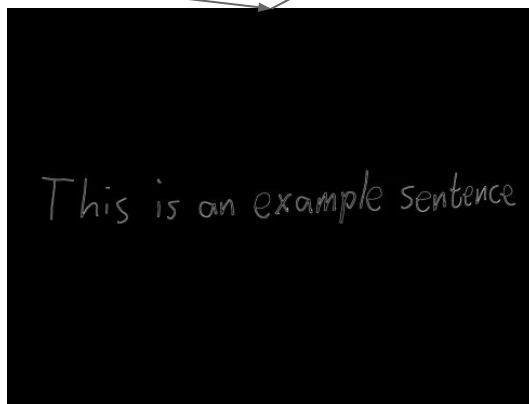
Image is sent
into the program

Contours are
identified



Individual
contours are
extracted for
image
manipulation

A final,
processed ima
is sent to th
CNN



RESULTS

- Trained Convolutional Neural Network on the EMNIST dataset to 88% testing accuracy

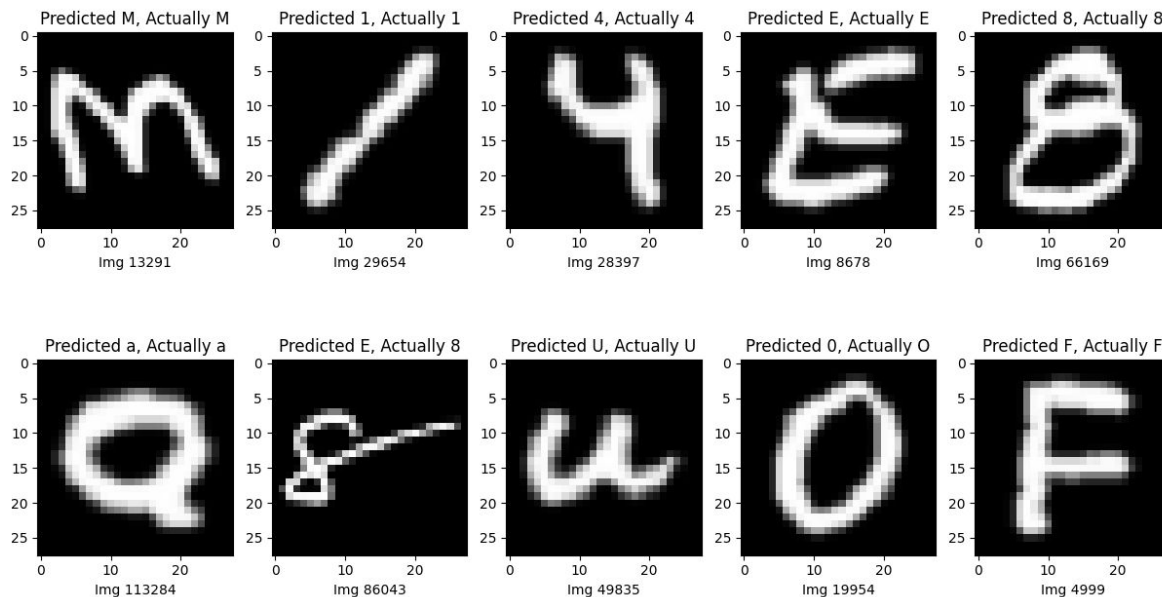


Photo of Program Outputs [5]

RESULTS (CONT.)

- Created program to extract characters to feed into model

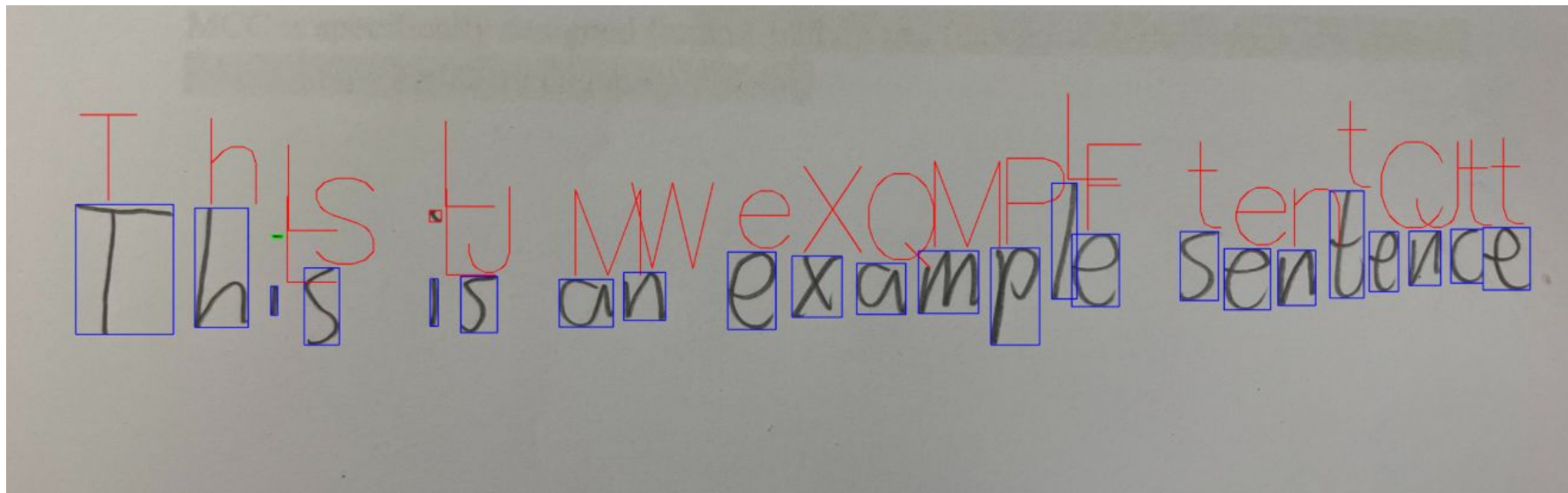
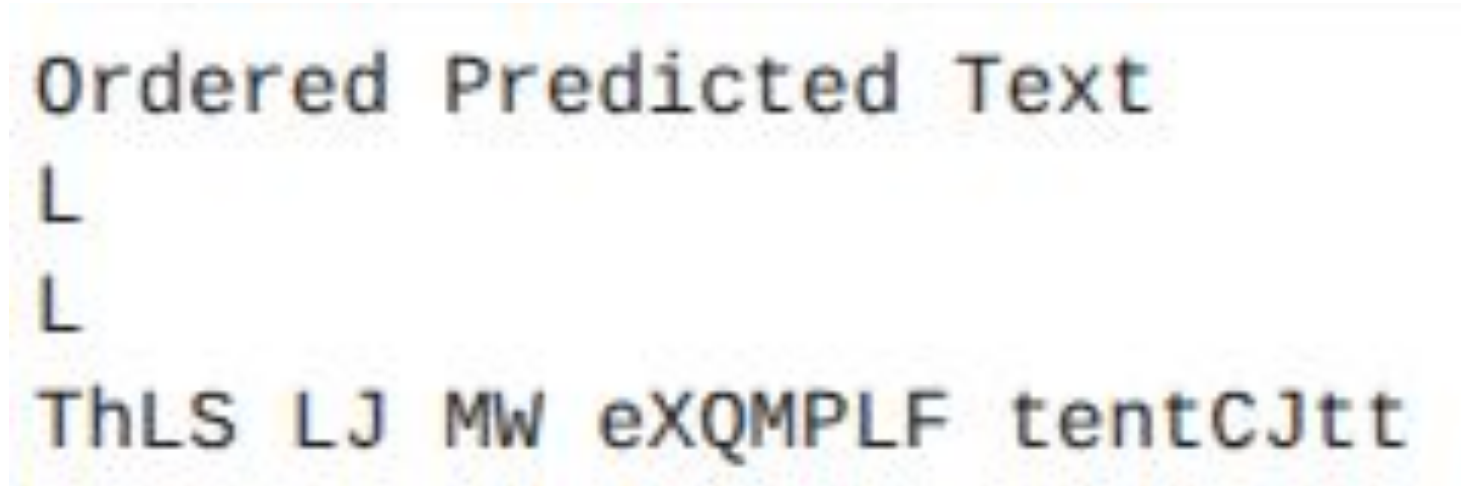


Photo of Program Outputs [5]

RESULTS (CONT.)

- Printed the output in lines it identified, maintaining format.



```
Ordered Predicted Text
L
L
ThLS LJ MW eXQMPLF tentCJtt
```

Photo of Program Outputs [5]

RESULTS (EXAMPLE) (12/5/25)

This is an example sentence

Using a lot of different

Letters and spacing

Ordered Predicted Text
2eCeL nnd PI L12IIb eeII52D5
IenIIe C L5I 2n d1L2ID5P2b
LeIIeI5 2Dd 5P2C1Inb

Photo of Program Outputs [5]

RESULTS (EXAMPLE) (3/6/25)

Image with Predicted Characters

The quick brown

Fox jumps over

The lazy dog

Ordered Predicted Text

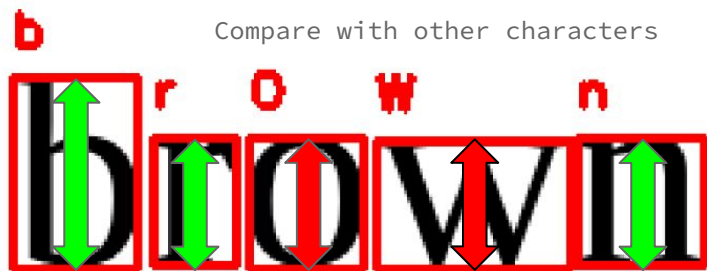
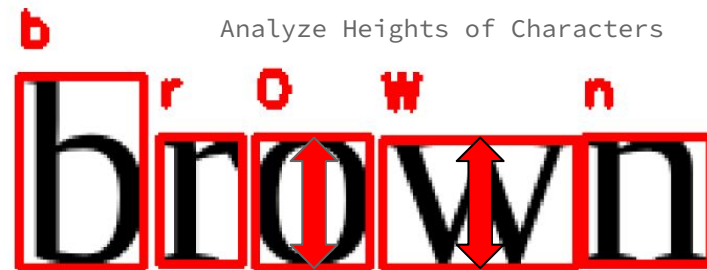
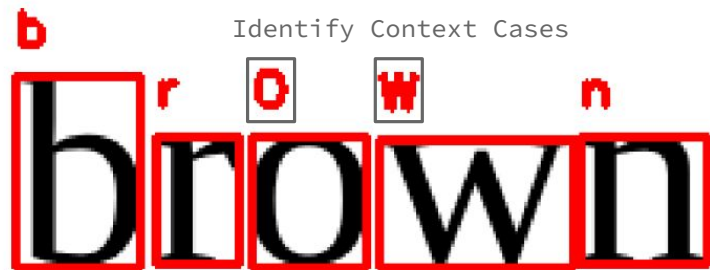
The qUICK brOWn

FOX JUMPS Over

The LaZY dOg

Photo of Program Outputs [5]

RESULTS (CONTEXT CASE EXAMPLE) (3/6/25)



Update Predictions Accordingly

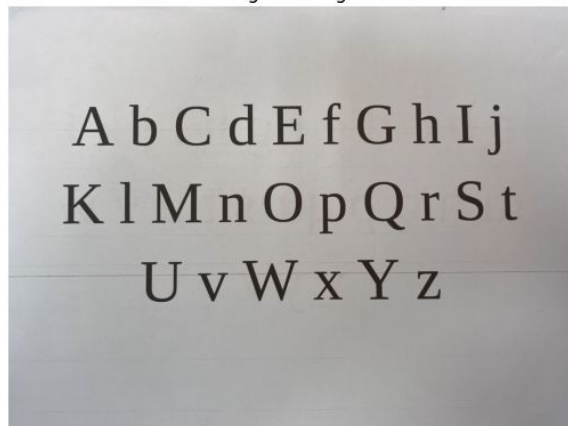
Final Updated Character Predictions (with refined size heuristic):
The quIck brown
fox JumPs over
The LazY dog

UPDATED CONTEXT CASE REFINEMENT

- Needed improved context case identification
- Size heuristic
 - Very basic
 - Works for some lowercase characters, while others aren't classified correctly
 - Ascenders and descenders
 - b, d, y, p, q, j, k, l, i, t, h, f, g
 - Identify ascenders and descenders, and use them to determine taller/shorter characters
 - Find baseline, or height of the middle of the line
- Contextual identification
 - If nearby characters are upper/lower, character in question is probably similar

RESULTS (ALTERNATE CONTEXT CASE EXAMPLE) (4/30/26)

Original Image



Original Image

Predicted characters:

A b C d E f G h I J
K L M n O P Q r S t
U V W X Y Z

Predicted Text
WITHOUT Context

Final Updated Character Predictions

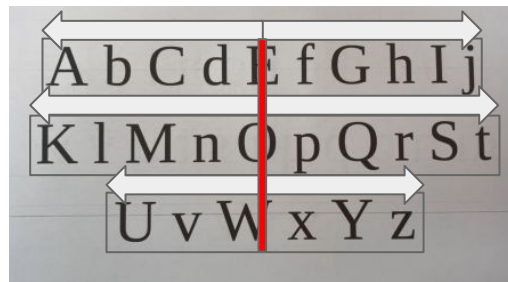
A b C d E f G h I j
K L M n O p Q r S t
U v W x Y z

Predicted Text
WITH Context

TEXTUAL ORGANIZATION RECOGNITION

- Not all images are left-centered
- Need to identify the organization of the text
- Analyze the cumulative line bounding boxes
- Find relation between each line
- Final output:

Detected alignments for each line:
['center', 'center', 'center']



Text is centered on middle line

Predicted characters:

A b C d E f G h I J
K L M n O P Q r S t
U V W X Y Z

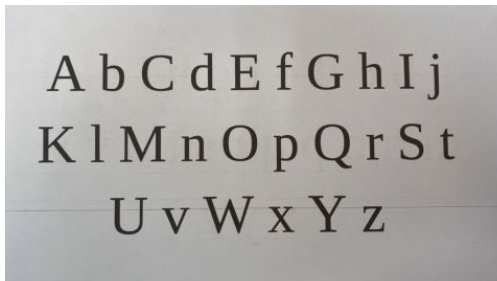
Text is left-centered on line

MARKDOWN FILE CONVERSION

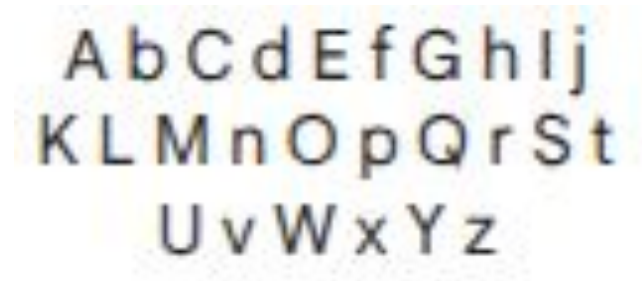
- Converted the output to a markdown file
- Previewing the markdown file shows the similarity between the original image and the generated output
- Markdown text:

```
<div align="center">AbCdEfGhIj</div>  
<div align="center">KLMnOpQrSt</div>  
<div align="center">UvWxYz</div>
```

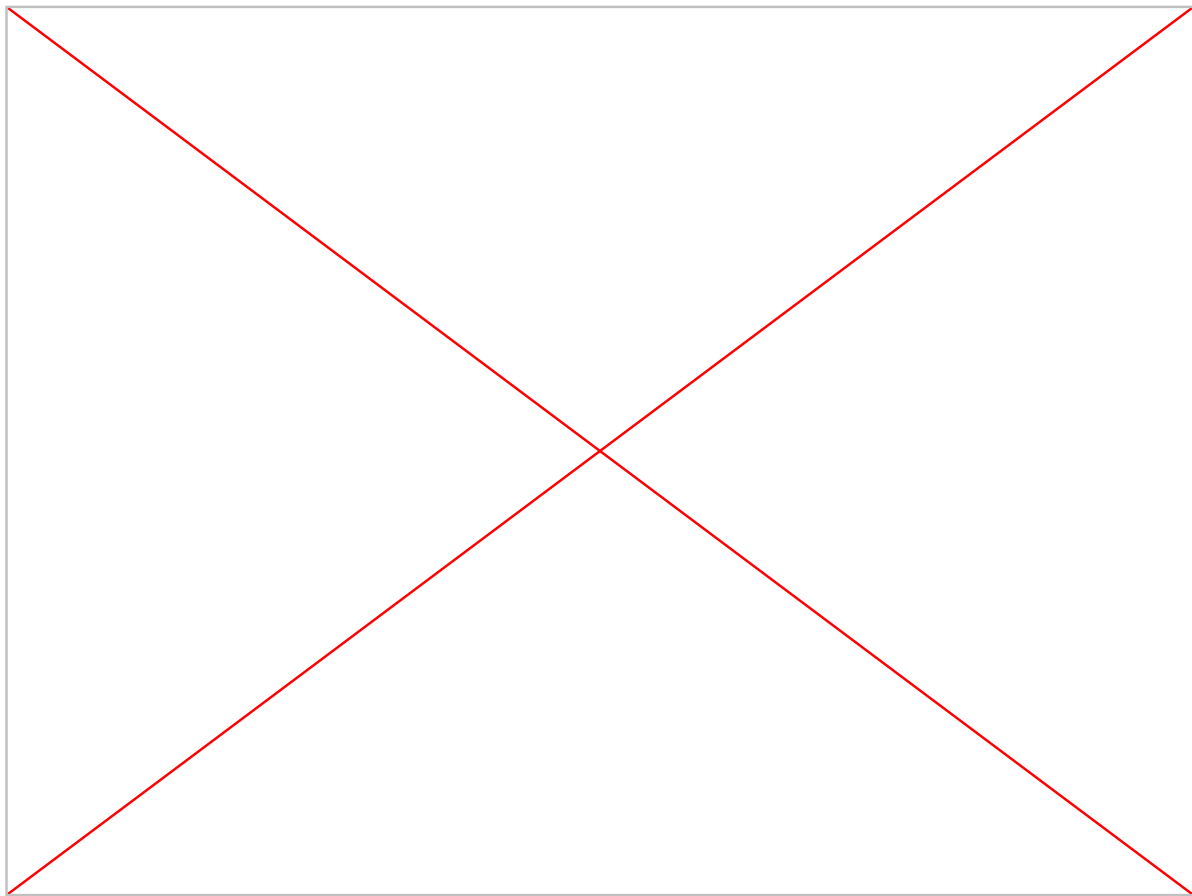
- Original Image:



- Final Markdown Output:



DEMO



Video Modeling Program Outputs [5]

LIMITATIONS

- Punctuation will be an issue, as the EMNIST dataset does not have any training images
- PDF file transformation may prove difficult as PDF is proprietary
 - This can be remedied by simply converting the finished product to another file format
- Context cases were not identified 100% accurately
 - Some characters, like 'l', are more difficult to classify, as they are very similar to their uppercase counterparts in both size and shape

CONCLUSION AND FUTURE WORK

- Able to create a model which accurately identifies alphanumeric characters
- Able to identify textual organization of the image
- Possible improvements include:
 - Identifying punctuation
 - Improving the CNN accuracy
 - Be able to identify structures like bulleted lists
 - Convert results to other file format like Word or PDF

REFERENCES

- [1] O. US EPA, “Identifying Greener Paper,” www.epa.gov, Nov. 14, 2014. Available: <https://www.epa.gov/greenerproducts/identifying-greener-paper>
- [2] P. Dizikes, How to Tackle the Global Deforestation Crisis. 2023. Accessed: May 20, 2026. [Online]. Available: <https://news.mit.edu/2023/tackling-global-deforestation-crisis-0919>
- [3] Cohen, G., Afshar, S., Tapson, J., and van Schaik, A., “EMNIST: an extension of MNIST to handwritten letters”, *arXiv e-prints*, Art. no. arXiv:1702.05373, 2017. doi:10.48550/arXiv.1702.05373.
- [4] “Pen to Print – Handwriting to Text Online OCR,” Pen-to-print.com, 2025. Available: <https://www.pen-to-print.com/>
- [5] B. DiCarlo, “Google Colab,” Google.com, 2019. Available: https://colab.research.google.com/drive/1PZP0kqVaEmtcIJ8A4_y981bLAS3kFw86#scrollTo=3TMvbNJw p8qZ. [Accessed: Oct. 17, 2025]

Q&A

THANKS!